

Auteur: Siebe Mees
 Studierichting: Informatica
 Oefeningenreeks 5G nr. 1.13

$$f(x) = \sqrt{x} \sqrt[4]{1-x} \text{ op } [0, 1]$$

x	0	$\frac{2}{3}$	1
$f(x)$	0	+	0
$f'(x)$	+	0	-
$f''(x)$	-	-	-
$f(x)$	0	$\nearrow M\left(\frac{\sqrt{2}}{3^{3/4}}\right)$	$\searrow$ 0
		$\cup$	$\cap$

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$$\begin{aligned}
 V &= \pi \int_0^1 (\sqrt{x} \sqrt[4]{1-x})^2 dx \\
 &= \pi \int_0^1 x \sqrt{1-x} dx \\
 t &= 1-x \quad (x = 1-t, \quad dx = -dt) \\
 &= \pi \int_1^0 (1-t) \sqrt{t} (-dt) \\
 &= \pi \int_0^1 (1-t) t^{1/2} dt \\
 &= \pi \int_0^1 (t^{1/2} - t^{3/2}) dt \\
 &= \pi \left[\frac{2}{3} t^{3/2} - \frac{2}{5} t^{5/2} \right]_0^1 \\
 &= \pi \left(\frac{2}{3} - \frac{2}{5} \right) \\
 &= \frac{4\pi}{15}
 \end{aligned}$$

References